

The Higgs Extension Classifier: Extra Scalars and Same-Scope Carrier Preservation

Paper 5 of the Higgs Structural Endpoint Arc

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Abstract

This paper classifies Higgs-domain extension claims after the Higgs bridge/fixation carrier, structural identity endpoint, scale-standing theorem, and vacuum-route-status theorem have already been fixed. It does not rule out beyond-Standard-Model scalar sectors, composite Higgs constructions, second doublets, portals, supersymmetric Higgs sectors, pseudo-Nambu-Goldstone presentations, dilaton/radion-like objects, or threshold updates. It denies only the primitive inference that an additional scalar-like, vev-like, coupled, composite, vacuum-active, or scale-active object is automatically a second same-scope Higgs identity.

The paper defines a model-neutral extension normal form

$$R_H^{\text{ext}} = (D_H, C_H, H_{\text{role}}, \Sigma_{\text{ext}}, \Pi_C, \Pi_{\text{id}}, \Pi_{\text{scale}}, \Pi_{\text{vac}}, \text{Cert}_{\text{ext}}, \rho_H^{\text{ext}})$$

and a layered extension classifier. The raw extension type records the submitted field or sector form; the carrier-relation layer records whether the submission preserves, enriches, duplicates, splits, replaces, adds to, or updates the carrier; the extension-disposition layer records whether same-scope use is carrier-indexed, downstream enrichment, role reassembly, richer-scope update, or failure; a separate burden layer records non-reducing same-scope replacement submissions as counterexample-burden submissions rather than standing survivors. The central theorem is not phenomenological exclusion. It is standing classification: every extension claim must be typed by its relation to C_H . Same-scope extra-Higgs identity has no standing-closed survivor outside C_H unless a discharged faithful same-domain counterexample preserves D_H , H_{role} , identity redescription, scale standing, and vacuum-route status without reducing to C_H . No such survivor is supplied by scalar similarity, vev participation, mixing, coupling, compositeness, hierarchy repair, thermal effect, or vacuum route availability alone.

Keywords: Higgs extension; extra scalar; composite Higgs; two-Higgs-doublet model; singlet scalar; portal sector; supersymmetry; pNGB Higgs; AASC; carrier preservation; structural identity; role splitting; same-scope; target update; standing closure.

Read this first: proof spine

The proof has five layers.

Layer I: inherited carrier and identity. Paper 1 establishes the local carrier package and Paper 2 fixes Higgs identity as admissible occupation of $C_H = H_{\text{br/fix}}$. Paper 3 deletes primitive scale standing and Paper 4 classifies vacuum claims as route statuses. Paper 5 does not rederive these modules; it classifies extension claims downstream of them.

Layer II: extension target and normal form. A submitted extension claim $\text{Ext}_H(x)$ is not evaluated by field content alone. It is evaluated by the tuple R_H^{ext} , which records the fixed domain, the carrier, the role bundle, the added sector, and the maps needed to preserve carrier identity, structural identity, scale status, and vacuum-route status.

Layer III: deterministic layered classification. Extension claims receive non-rival classifications: raw type, boundary/update decision, carrier relation, standing disposition, and, where replacement is asserted, counterexample-burden classifier. A singlet, second doublet, composite sector, portal, or supersymmetric sector is a raw extension type. Its standing is decided only after the same-scope boundary, carrier relation, and certificates are declared.

Layer IV: no primitive extra-Higgs identity. Scalar similarity, vev participation, mixing, coupling to C_H , compositeness, role-splitting, vacuum effect, and hierarchy effect do not generate same-scope Higgs identity. They either enrich or describe C_H , add non-Higgs field content, reassemble into C_H , declare a target update, or fail as primitive identity grounds.

Layer V: faithful counterexample burden. A submitted same-scope extra-Higgs counterexample must preserve D_H , H_{role} , C_H , the Paper 2 identity endpoint, the Paper 3 scale-standing boundary, and the Paper 4 route-status boundary while refusing reduction to C_H . Such a submission is a counterexample burden, not a standing survivor, until every preservation and non-reduction condition is discharged.

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1 Introduction: extensions as standing claims

Higgs extensions are normally introduced by field content: an extra singlet, a second doublet, a composite bound state, a pNGB realization, a supersymmetric sector, a portal, a radion-like scalar, or a threshold structure. That language is physically useful. It is not by itself a same-scope identity claim. A new scalar may exist in a model without being a new Higgs identity. A new sector may alter vacuum behavior without replacing the carrier. A composite description may preserve the same Higgs role under a different representation or may change the target. A role-splitting construction may distribute functions across several formal objects while reassembling into one carrier role.

This paper therefore treats extensions as standing claims. The question is not whether a sector is interesting, calculable, measurable, or model-building relevant. The question is what kind of Higgs-domain standing claim it makes. Does it preserve C_H ? Does it enrich descriptions or dynamics while leaving identity fixed? Does it duplicate a representative? Does it split the carrier role and then reassemble it? Does it add a field that is not a Higgs identity? Does it change the target domain? Or does it assert primitive Higgs identity without the required carrier preservation?

The result is a classifier, not a no-BSM theorem. The paper does not rule out extra scalar physics, composite Higgs programs, supersymmetric Higgs sectors, two-Higgs-doublet models, Higgs portals, pNGB realizations, or richer electroweak-sector mechanisms. It rules out only an inferential shortcut:

$$\text{additional scalar-like structure} \implies \text{second same-scope Higgs identity.}$$

Same-scope Higgs identity is already fixed by C_H . Extension standing must be assigned relative to that fact.

2 Dependency lock and inherited package

This manuscript works downstream of the first four papers of the Higgs structural endpoint arc. It imports their local outputs rather than the full kernel proof. The dependency posture is compact because Paper 1 carries the full AASC kernel and local Higgs carrier specialization.

Table 1: Inherited modules used by Paper 5.

Module	Imported object	Use in this paper
Paper 1	$D_H, C_H, H_{\text{role}}, R_H, \mathbf{SF}_H$	Fixes the carrier package and the same-scope domain in which extension claims are evaluated.
Paper 2	$\text{Id}_H = \text{Occ}_{D_H}(C_H)$	Blocks representative or scalar similarity from becoming identity ground without carrier preservation.
Paper 3	R_H^{scale} and no primitive Higgs scale	Prevents extension mass, vev, hierarchy, or running claims from grounding identity.
Paper 4	R_H^{vac} , route availability, standing closure	Prevents vacuum, lower-branch, thermal, or EWBG effects from becoming primitive extension identity.

Module	Imported object	Use in this paper
Fixed-domain support	no fifth governance-bearing locus	Used to close the extension-function classifier.
Impossibility support	no independent standing generators, no cross-domain standing transfer, no same-act repair	Blocks extension success, usefulness, or route availability from generating same-scope Higgs identity.

Definition 2.1 (Inherited Higgs-domain package). The inherited package is

$$\mathcal{P}_H = (D_H, C_H, H_{\text{role}}, R_H, R_H^{\text{id}}, R_H^{\text{scale}}, R_H^{\text{vac}}, \text{SF}_H).$$

It fixes the domain, carrier, identity normal form, scale normal form, vacuum route normal form, and standing-filter discipline available to extension claims.

Nonclaim 2.2 (No phenomenological survey). The paper does not survey all BSM Higgs models, compute cross sections, predict masses, select a scalar potential, derive coupling values, or test experimental exclusions. The paper classifies the standing role of extension claims relative to $\mathcal{P}_H$.

Nonclaim 2.3 (No anti-extension result). The paper does not deny extra scalar sectors, compositeness, supersymmetry, portals, pNGB mechanisms, or richer electroweak physics. It distinguishes additional physics from same-scope Higgs identity.

3 Extension claim taxonomy

Definition 3.1 (Higgs-domain extension claim). A Higgs-domain extension claim, written $\text{Ext}_H(x)$, is a submitted claim that added structure x changes, preserves, enriches, duplicates, splits, replaces, or updates the Higgs-domain carrier, identity, scale, or vacuum route status.

Definition 3.2 (Raw extension type). The raw extension type $\rho_{\text{type}}^{\text{ext}}(x)$ records what kind of formal or physical addition is submitted before standing evaluation. The admissible raw-type labels used here are:

ExtraScalar, Singlet, SecondDoublet,
Composite, pNGB, DilatonRadion,
SUSYSector, PortalSector, RoleSplitSector,
UVThreshold.

Raw type is descriptive. It does not determine Higgs identity.

Table 2: Raw extension taxonomy.

Raw type	What it submits	Immediate standing caution
Extra scalar	A scalar-like carrier or excitation associated with the Higgs sector.	Scalarity alone is not Higgs identity.
Singlet extension	A gauge-singlet scalar with portal or mixing structure.	Coupling to C_H is not occupation of C_H .

Raw type	What it submits	Immediate standing caution
Second doublet	An additional doublet or enlarged scalar multiplet.	Additional representation content is not automatically second identity.
Composite Higgs	A bound-state, condensate, or substructure presentation.	A representation of the carrier may preserve identity or may shift scope.
pNGB route	Higgs-like object as a pseudo-Nambu-Goldstone mode.	Protection or origin story is not identity unless the carrier role is preserved.
Dilaton/radion-like object	Scalar linked to scale, conformal, or geometric sector.	Scale role or metric projection is downstream of C_H .
SUSY sector	Extended Higgs sector, partners, or supersymmetric threshold.	Richer field inventory normally changes scope unless same carrier preservation is certified.
Portal sector	Hidden or dark sector coupled to Higgs operators.	Portal coupling may enrich dynamics without changing Higgs identity.
Role-split sector	Formal division of bridge functions across multiple objects.	Splitting must reassemble into the same role or declare target update.
UV/threshold sector	New high-scale boundary, threshold, or completion.	High-scale repair is a target update unless same-scope invariants are preserved.

4 Extension normal form

A standing-bearing extension claim must declare more than field content. It must declare how added structure relates to the carrier, identity endpoint, scale status, and vacuum route status already fixed.

Definition 4.1 (Extension normal form). The extension normal form is

$$R_H^{\text{ext}} = (D_H, C_H, H_{\text{role}}, \Sigma_{\text{ext}}, \Pi_C, \Pi_{\text{id}}, \Pi_{\text{scale}}, \Pi_{\text{vac}}, \text{Cert}_{\text{ext}}, \rho_H^{\text{ext}}),$$

where:

- Σ_{ext} is the submitted added carrier, field, threshold, source, or sector content;
- Π_C is the carrier-preservation map;
- Π_{id} is the identity-redescription map relative to Paper 2;
- Π_{scale} is the scale-status map relative to Paper 3;
- Π_{vac} is the vacuum/thermal route-status map relative to Paper 4;
- Cert_{ext} is the certificate declaring same-scope preservation, enrichment, target update, or failure;
- ρ_H^{ext} is the layered extension classifier.

Definition 4.2 (Same-scope extension normalizer). For a submitted $\text{Ext}_H(x)$, the same-scope partial normalizer $\text{NF}_{H,\text{ext}}^{\text{same}}(x)$ is defined exactly when x declares the fields of R_H^{ext} , preserves the inherited

domain D_H , preserves C_H or a role-equivalent redescription of it, and supplies the maps required for identity, scale, and vacuum-route discipline.

Definition 4.3 (Target-update extension normalizer). For a submitted $\text{Ext}_H(x)$, the update partial normalizer $\text{NF}_{H,\text{ext}}^{\text{update}}(x)$ is defined exactly when x declares an update to $D_{H,+}$, identifies the changed carrier inventory or invariant bundle, and supplies a descent or comparison map back to the Higgs-domain target being discussed. It is not a same-scope normalizer and cannot by itself certify same-scope survival.

Theorem 4.4 (Extension normal-form necessity). *If an extension claim is same-scope standing-bearing, then $\text{NF}_{H,\text{ext}}^{\text{same}}(x) \downarrow$. If an extension claim changes the Higgs-domain target, then $\text{NF}_{H,\text{ext}}^{\text{update}}(x) \downarrow$. A single submitted claim may not use $\text{NF}_{H,\text{ext}}^{\text{update}}$ as proof of same-scope survival.*

Proof. A same-scope standing-bearing extension claim preserves Higgs-domain standing. By Papers 1 and 2, same-scope identity standing is carrier-indexed by C_H . By Paper 3, scale claims require R_H^{scale} . By Paper 4, vacuum and thermal claims require R_H^{vac} and route-status classification. Therefore same-scope use must declare the added sector, the carrier-preservation map, identity redescription, scale import, vacuum/thermal import, and certificate while preserving D_H ; these data define $\text{NF}_{H,\text{ext}}^{\text{same}}$. If the added structure changes the carrier inventory, invariant bundle, admissibility boundary, comparison target, or source domain, the claim is no longer same-scope. It may still be an admissible richer-scope construction, but only by declaring $D_{H,+}$ and the associated descent data, which define $\text{NF}_{H,\text{ext}}^{\text{update}}$. Treating the update normalizer as same-scope survival would erase the fixed-domain distinction established by the inherited package. $\square$

5 Layered extension classifier

Extension classification must be layered. Raw extension type, boundary/update status, carrier relation, standing disposition, and counterexample burden are not rival categories. A second doublet may be a raw type, a carrier-preserving enrichment in carrier relation, and a richer-scope update in standing disposition depending on the declared certificate.

Definition 5.1 (Carrier-relation layer). The carrier-relation layer $\rho_{\text{carrier}}^{\text{ext}}(x)$ takes one of the following values:

{SameCarrier, CarrierPreservingEnrichment, RepresentativeDuplication, RoleSplitCandidate,
ReplacementClaim, AdditionalField, TargetUpdate, FailedPrimitive}.

Definition 5.2 (Standing-disposition and burden layers). The standing disposition $\rho_{\text{stand}}^{\text{ext}}(x)$ takes one of the following values:

{CarrierIndexedExtension, AdditionalNonHiggsField, RicherScopeUpdate,
RoleSplitReassembly, FailedExtraHiggsIdentity}.

A replacement claim that purports to preserve D_H while refusing reduction to C_H receives the separate burden status

$$\rho_{\text{burden}}^{\text{ext}}(x) = \text{CounterexampleSubmissionPending}.$$

This burden status is not a standing survivor. It records that the submitted claim has entered the faithful-counterexample test without satisfying it.

Table 3: Carrier-relation layer.

Carrier relation	Meaning	Standing consequence
SameCarrier	The added presentation is another representation of C_H .	Same Higgs identity under redescription.
Carrier-preserving enrichment	Added structure changes dynamics or descriptions while leaving C_H primitive.	Downstream enrichment, not new identity.
Representative duplication	The extension duplicates a vev, scalar, mass, mixing, or field description.	Representative-only unless carrier role changes.
Role-split candidate	H_{role} functions are distributed across multiple objects.	Must reassemble into C_H , shift scope, or fail.
Replacement claim	The extension claims to replace C_H while remaining same-scope.	Enters the counterexample-burden layer unless it reduces to same carrier or fails.
Additional field	Added object exists or is modeled without occupying Higgs identity.	Additional non-Higgs field.
Target update	Carrier inventory, invariant bundle, or comparison target changes.	Richer-scope update.
Failed primitive	Primitive Higgs identity is asserted from scalarity, vev, coupling, or success.	Failed extra-Higgs identity.

Definition 5.3 (Extension decision procedure). The extension classifier is assigned by the following ordered decision procedure.

1. **Boundary check.** If the claim changes carrier inventory, invariant bundle, admissibility boundary, comparison target, or source domain, it is classified by $\text{NF}_{H,\text{ext}}^{\text{update}}$ as **RicherScopeUpdate**. It does not proceed as a same-scope survivor.
2. **Same-scope preservation check.** If the claim remains in D_H , it is tested against the same-scope preservation criteria of Section 6. Failure without an update certificate yields **FailedExtraHiggsIdentity** when identity standing is asserted.
3. **Carrier-relation check.** A preserving claim is assigned its carrier relation: same carrier, carrier-preserving enrichment, representative duplication, role-split candidate, replacement claim, additional field, or failed primitive. These labels are non-rival with raw field type.
4. **Standing-disposition check.** Same carrier and carrier-preserving enrichment yield **CarrierIndexedExtension**; representative duplication or additional field yields **AdditionalNonHiggsField** unless identity is primitively asserted, in which case it fails; role splitting yields **RoleSplitReassembly**, update, or failure; primitive representative authority yields **FailedExtraHiggsIdentity**.
5. **Counterexample-burden check.** A replacement claim that passes the same-scope preservation check but refuses reduction to C_H is assigned $\rho_{\text{burden}}^{\text{ext}}(x) = \text{CounterexampleSubmissionPending}$. This records an undischarged proof burden, not a standing disposition.

Theorem 5.4 (Extension classifier totality). *Every submitted $\text{Ext}_H(x)$ receives exactly one raw-type*

classification and one boundary decision. Every complete same-scope claim receives exactly one carrier-relation classification and exactly one standing disposition. Non-reducing replacement submissions receive a separate counterexample-burden classifier rather than an additional standing disposition.

Proof. Raw type records the submitted descriptive form and is total for the catalogue under evaluation; unlisted forms enter as $UV_{\text{Threshold}}$ or by explicit update declaration. The boundary check is first: a target-changing claim is evaluated by $NF_{H,\text{ext}}^{\text{update}}$, not by same-scope preservation. A non-updating claim enters the same-scope preservation check; failure there cannot become a same-scope survivor. A preserving claim then receives a carrier-relation label that identifies whether the submitted structure is same carrier, enrichment, representative duplication, role split, replacement, additional field, or failed primitive authority. The standing-disposition layer is determined from that carrier relation and certificate. A replacement claim that refuses reduction to C_H is not given an additional standing disposition; it is registered as a counterexample-burden submission. A second standing disposition would either change no standing-bearing fact and be bookkeeping, or change carrier/target status and be a different carrier-relation classification. $\square$

6 Same-scope preservation tests

A same-scope extension is not identified by sharing a scalar notation or coupling to the observed Higgs sector. It is identified by preserving the Higgs-domain role package.

Definition 6.1 (Same-scope extension preservation test). A submitted x passes the same-scope extension preservation test if all of the following hold:

1. **Gauge inheritance:** the extension preserves the gauge/representation inheritance relevant to H_{role} .
2. **Mass-response preservation:** it preserves admissible mass-response readiness without importing mass values as identity grounds.
3. **Deformation stability:** it preserves the deformation-response role associated with C_H .
4. **Identity continuity:** it preserves $\text{Id}_H = \text{Occ}_{D_H}(C_H)$ under the inherited identity-redescription discipline.
5. **Standing transfer discipline:** it does not transfer standing from empirical success, calculability, route availability, or external domain authority.
6. **Scale discipline:** scale, threshold, mass, or hierarchy claims factor through R_H^{scale} .
7. **Vacuum-route discipline:** vacuum, thermal, and baryogenesis claims factor through R_H^{vac} .
8. **No hidden fifth classifier:** no additional same-scope identity classifier is introduced after carrier identity has closed.

Table 4: Preservation tests and failure modes.

Test	Failure mode	Allowed result
Gauge inheritance	New representation content breaks inherited order rather than preserving it.	Target update or failed same-scope claim.

Test	Failure mode	Allowed result
Mass-response	Mass matrix, Yukawa topology, or fitting success selects identity.	Downstream route or standing-smuggling failure.
Deformation stability	The extension changes deformation role without declaring carrier update.	Target update.
Identity continuity	New object is called Higgs by scalar similarity alone.	Failed extra-Higgs identity.
Standing transfer	Success, usefulness, calculability, or measurement is treated as authorizing identity.	Failed primitive or additional non-Higgs field.
Scale discipline	Scale/hierarchy rescue is used as identity ground.	Carrier-indexed scale route or target update.
Vacuum-route discipline	Vacuum stabilization or first-order transition is treated as Higgs identity.	Vacuum route status or target update.
No fifth classifier	Extra identity criterion is introduced without preserving C_H .	Fixed-domain failure.

Lemma 6.2 (Preservation-test necessity). *A same-scope extension that fails any preservation test cannot be a same-scope Higgs identity survivor. A richer certificate may define an update, but it does not pass the same-scope preservation test.*

Proof. The tests are the local expression of the inherited package. Failing gauge inheritance, mass-response preservation, deformation stability, identity continuity, scale discipline, or vacuum-route discipline breaks one of the fixed role components used by Papers 1–4. Failing standing transfer discipline or no-fifth-classifier discipline imports a new authorizer into the fixed domain. In either case the extension does not remain a same-scope identity survivor. A richer certificate can be meaningful, but then the claim is evaluated by $\text{NF}_{H,\text{ext}}^{\text{update}}$ in $D_{H,+}$, not as a same-scope survivor in D_H . $\square$

7 Extra scalar non-identity

The simplest extension error is the inference from scalarity to identity. Paper 2 already blocks scalarity as primitive Higgs identity. Paper 5 adds the extension-specific form: a second scalar-like object does not become a second Higgs identity by being scalar-like, vev-like, mixed, coupled, or nearby in the potential.

Theorem 7.1 (Extra scalar non-identity). *For any submitted extra scalar s ,*

$$\text{Ext}_H(s) \wedge \text{ScalarLike}(s) \not\Rightarrow \text{Id}_H(s).$$

More specifically, scalar similarity, vev participation, coupling to C_H , mass mixing, potential participation, and phenomenological relevance do not supply primitive Higgs identity standing.

Proof. Scalarity is a representative feature, not the role bundle. Vev participation is a vacuum or symmetry-reduction description and is not identity ground without C_H . Coupling and mixing are dynamical or measurement-relevant relations, not carrier occupation. Potential participation is a chart or model description unless it descends to a carrier certificate. Phenomenological relevance is success or usefulness and cannot generate standing. Therefore an extra scalar may be an additional field, enrichment, representative duplication, target update, or counterexample submission pending, but it is not a second same-scope Higgs identity merely by possessing these features. $\square$

Corollary 7.2 (Extra scalar outcome). *An extra scalar has one of the following extension outcomes: carrier-preserving enrichment, representative duplication, additional non-Higgs field, target update, failed primitive, or same-scope counterexample submission pending. It is not automatically an extra Higgs identity.*

8 Carrier-preserving enrichment

Definition 8.1 (Carrier-preserving enrichment). A submitted extension x is a carrier-preserving enrichment when x adds dynamics, interaction channels, symmetry structure, measurement signatures, or route content while leaving C_H as the primitive Higgs identity carrier.

Lemma 8.2 (Carrier-preserving enrichment). *If $\text{Pres}_C(x)$ holds and the added structure does not alter D_H , H_{role} , R_H^{id} , R_H^{scale} , or R_H^{vac} , then x is downstream enrichment and not a new primitive Higgs identity.*

Proof. Preserving C_H preserves the identity carrier. If the inherited identity, scale, and vacuum route modules remain unchanged, the extension changes downstream descriptions or dynamics rather than the primitive identity ground. Thus the proper classification is carrier-indexed extension or additional route content, not second Higgs identity. $\square$

Remark 8.3 (Why enrichment matters). This classification is not dismissive. Carrier-preserving enrichment may affect phenomenology, stability, phase transition structure, dark-sector coupling, or collider signatures. The point is that such effects do not by themselves alter the identity carrier.

9 Composite and pNGB scope tests

Composite and pNGB presentations are often more subtle because they can claim not merely extra field content, but a different underlying realization of the Higgs role.

Definition 9.1 (Composite preservation alternatives). A composite or pNGB claim has one of three preservation alternatives:

1. **representational preservation:** the composite presentation realizes the same C_H under a different representation;
2. **richer-scope construction:** the composite dynamics changes carrier inventory or invariant bundle and declares $D_{H,+}$;
3. **identity failure:** the claim neither preserves C_H nor declares a target update.

Theorem 9.2 (Composite scope test). *Every composite or pNGB Higgs claim is classified as same-carrier representation, carrier-preserving enrichment, target update, or failed primitive. No composite claim supplies same-scope Higgs identity merely by replacing an elementary-field presentation.*

Proof. If the composite presentation preserves the role bundle and identity-redescription map, then it is a representation or realization of C_H , not a second same-scope identity. If it changes the underlying carrier inventory, source structure, symmetry protection, or admissibility boundary, then it is a richer-scope construction. If it does neither, it lacks identity continuity. Elementarity or compositeness is therefore not the identity criterion; preservation of C_H is. $\square$

10 Role-splitting analysis

Some extension claims distribute Higgs-domain functions across multiple formal objects: one object provides symmetry breaking, another mass response, another stabilization, and another phenomenological scalar. The classifier must avoid two errors: falsely rejecting genuine role-distributed theories and falsely accepting any role distribution as second Higgs identity.

Definition 10.1 (Role split). A role split occurs when functions in H_{role} are distributed across multiple submitted objects $x_1, \dots, x_n$ instead of being presented by a single formal representative.

Definition 10.2 (Role-split reassembly). A role split reassembles into C_H when the distributed package preserves the same D_H , same H_{role} , same identity-redescription class, and same admissibility boundary as C_H .

Theorem 10.3 (Role-splitting exhaustion). *A role-splitting extension has exactly one standing disposition: it reassembles into C_H , declares a richer-scope target update, or fails identity continuity.*

Proof. If the distributed roles preserve the same domain, role bundle, identity-redescription relation, and admissibility boundary, then the distribution is a presentation of the same carrier role. It reassembles into C_H . If preserving the distributed package requires new carrier inventory, a new invariant bundle, or a new admissibility boundary, then the proposal is a target update. If neither reassembly nor update is declared, the distributed objects fail to supply same-scope identity continuity. These three cases exhaust the possibilities: same role, changed role/domain, or failure. $\square$

Corollary 10.4 (No role-count shortcut). *Multiple objects can be required by a richer target without being multiple same-scope Higgs identities. Conversely, a single object can fail identity if it does not preserve H_{role} . Cardinality of displayed fields is not the identity criterion.*

11 Scale and vacuum imports in extensions

Many extension proposals are motivated by hierarchy repair, stabilization of the Higgs potential, first-order transition generation, baryogenesis, or high-scale completion. These are legitimate motivations; they are not identity grounds.

Lemma 11.1 (Scale import discipline). *If an extension claim uses mass scale, threshold, hierarchy, running, naturalness, or UV sensitivity as a reason that x is a Higgs identity, then the scale portion must factor through R_H^{scale} . If it does not, the claim is standing-smuggling or target update.*

Proof. Paper 3 deletes primitive Higgs scale standing. Therefore scale features can only be carrier-indexed, measurement representative, transport route, boundary convention, metric projection, or target update. None of these independently supplies identity. An extension using scale behavior as identity must either show carrier preservation or change target; otherwise it smuggles scale standing into identity standing. $\square$

Lemma 11.2 (Vacuum import discipline). *If an extension claim uses stabilization, lower-branch repair, thermal transition, first-order EWPT, or EWBG success as a reason that x is a Higgs identity, then the vacuum/thermal portion must factor through R_H^{vac} . Route availability or route success does not itself create same-scope identity.*

Proof. Paper 4 separates vacuum representatives, route availability, and standing closure. A vacuum or thermal effect may be physically significant, but it is downstream of C_H . If an extension changes the finite-temperature target or supplies a new carrier, it is a target update. If it preserves C_H , it is carrier-preserving enrichment. It is not a primitive identity ground. $\square$

12 Unknown attribute classifier

Paper 5 is the arc’s formal bridge to unknown Higgs attributes. It does not predict attribute values. It determines what status a proposed attribute can have.

Definition 12.1 (Unknown Higgs attribute claim). An unknown Higgs attribute claim is a submitted claim that some not-yet-fixed property a belongs to the Higgs domain and bears identity, scale, vacuum, extension, or measurement significance.

Table 5: Unknown-attribute standing classifier.

Attribute mode	Standing route	Result
Measured coupling or width	Empirical representative of C_H .	Measurement content, not identity ground.
Rare decay or branching channel	Phenomenological route preserving C_H .	Carrier-preserving enrichment if same-scope.
Extra resonance	Additional field or representative duplication.	Not Higgs identity unless carrier preservation is certified.
Vacuum-stability effect	R_H^{vac} .	Route status or target update.
Thermal transition effect	R_H^{vac} and thermal target.	Route status or richer-scope update.
Hierarchy repair	R_H^{scale} .	Carrier-indexed scale content or target update.
Composite substructure	Π_{id} and Π_C .	Same carrier, scope update, or failure.
Role distribution	Role-split reassembly test.	Reassembles, updates target, or fails.

Theorem 12.2 (Unknown attribute classification). *Every Higgs-domain unknown-attribute claim is measurement representative, carrier-preserving enrichment, scale-status claim, vacuum-route claim, target update, role-split claim, or failed primitive identity claim.*

Proof. An attribute either describes an already fixed carrier, changes a route or measurement associated with that carrier, affects scale, affects vacuum/thermal status, alters the domain, distributes the carrier roles, or asserts primitive identity. These are exactly the registered categories supplied by Papers 1–4

and the extension classifier. A residual category would need to change Higgs-domain standing while preserving carrier, identity, scale, vacuum, and extension status; fixed-domain exhaustion excludes such a locus. $\square$

13 Extension outcome matrix

Table 6: Extension outcome matrix.

Claim type	Carrier relation	Required certificate	Allowed standing
Singlet scalar	additional field or enrichment	preserve C_H , declare mixing/coupling status	non-Higgs field or carrier-preserving enrichment
Second doublet	enrichment, role split, or update	representation, gauge, identity, and role-preservation certificate	enriched same carrier, role reassembly, or $D_{H,+}$
Composite Higgs	same carrier or target update	identity-redescription map and carrier-preservation map	representation of C_H or richer-scope construction
pNGB route	same carrier, enrichment, or update	protection mechanism plus carrier-preservation certificate	same identity under representation or richer target
SUSY Higgs sector	target update unless preservation certified	extra carrier inventory and descent map	richer-scope update or carrier-preserving enrichment
Portal sector	enrichment or additional field	portal status and no identity transfer	downstream route, not second identity
Dilaton/radion-like	scale projection or target update	scale-normal and carrier-preservation certificate	scale status, added field, or richer target
Role-split sector	role split	reassembly certificate or target-update declaration	reassembles into C_H , updates, or fails
UV threshold	target update or boundary convention	invariant-bundle and descent certificate	richer-scope update or scale route

14 Extension function exhaustion

The classifier is structural rather than a model inventory. It exhausts the functions an extension can perform relative to Higgs-domain standing.

Table 7: Extension functions and fixed-domain loci.

Extension function	Fixed-domain locus	Classifier result
Describe	lawful quotient / representative skin	representative duplication or same-carrier description
Add field content	carrier inventory boundary	additional field or target update
Enrich dynamics	admissible continuation while preserving carrier	carrier-preserving enrichment
Split roles	role-bundle preservation	reassembly, target update, or failure
Replace carrier	identity-redescription and carrier relation	same carrier, counterexample submission pending, or failure
Alter scale	scale-standing normal form	R_H^{scale} status or target update
Alter vacuum/thermal route	route-status normal form	R_H^{vac} status or target update
Select identity from success	standing partition	standing smuggling / failed primitive
Change domain	boundary typing/invariant bundle	richer-scope update

Lemma 14.1 (Extension-function exhaustion). *Any same-scope extension claim that is admissibility-bearing must perform one of the functions in the extension-function table. No further same-scope extension function remains.*

Proof. An extension can only affect Higgs-domain standing by describing the carrier, adding content, enriching continuation, distributing roles, replacing the carrier, altering scale status, altering vacuum/thermal route status, selecting identity from a representative or success criterion, or changing the domain. These correspond to lawful quotient, carrier inventory, admissible continuation, role bundle, identity redescription, scale normal form, vacuum normal form, standing partition, and boundary typing. A proposed further function would need to alter Higgs-domain standing while changing none of these loci. That is excluded by fixed-domain exhaustion. $\square$

15 Extension classification theorem

Theorem 15.1 (Extension standing filter). *For every submitted $\text{Ext}_H(x)$,*

$$\text{Ext}_H(x) \Rightarrow \rho_H^{\text{ext}}(x) \in \{\text{CarrierIndexedExtension}, \text{AdditionalNonHiggsField}, \text{RicherScopeUpdate}, \text{RoleSplitReassembly}, \text{FailedExtraHiggsIdentity}\}$$

when the claim reaches the extension-disposition layer. A non-reducing same-scope replacement submission is recorded separately as $\rho_{\text{burden}}^{\text{ext}}(x) = \text{CounterexampleSubmissionPending}$, which is not a standing survivor.

Proof. By extension-function exhaustion, an extension either preserves the carrier and remains carrier-indexed, adds non-Higgs field content, changes the target, reassembles a role split, or fails primitive

identity standing. These outcomes cover the possible standing-bearing effects of an extension after raw type and carrier relation have been separated. A replacement claim that purports to preserve the same scope while refusing reduction to C_H does not become a sixth standing disposition. It is registered as a counterexample-burden submission until the faithful counterexample conditions are discharged. Because the disposition layer is assigned after carrier relation, descriptive field type cannot decide the result. $\square$

Theorem 15.2 (Extension classification theorem). *Every Higgs-domain extension claim has a normal-form classification through R_H^{ext} . Claims that do not provide the normal-form fields are incomplete as standing claims. Claims that provide the fields are classified as carrier-indexed extension, additional non-Higgs field, richer-scope update, role-split reassembly, or failed extra-Higgs identity; non-reducing replacement submissions receive the separate pending-burden label `CounterexampleSubmissionPending`.*

Proof. The extension normal-form necessity theorem shows that same-scope standing-bearing claims require $\text{NF}_{H,\text{ext}}^{\text{same}}$ and target-updating claims require $\text{NF}_{H,\text{ext}}^{\text{update}}$. The layered classifier gives raw type, boundary decision, carrier relation, and extension outcome. The extension standing filter supplies the outcome classes, while the counterexample-burden layer records replacement submissions that have not discharged their burden and therefore are not same-scope standing survivors. Therefore every extension claim either lacks the information needed for standing evaluation, is evaluated as a target update, or receives one of the registered same-scope outcomes plus any applicable burden label. $\square$

16 Same-scope extra-Higgs burden and primitive deletion

Definition 16.1 (Same-scope extra-Higgs identity). A same-scope extra-Higgs identity is a submitted object x such that $x \neq C_H$, $\text{SameScope}_{D_H}(x)$, x preserves D_H and H_{role} , and x claims primitive Higgs identity standing without reducing to C_H under identity-preserving redescription.

Theorem 16.2 (Same-scope extra-Higgs counterexample burden). *A submitted same-scope extra-Higgs identity outside C_H is not a standing survivor merely by submission. It remains a counterexample-burden submission unless it preserves D_H , H_{role} , R_H^{id} , R_H^{scale} , and R_H^{vac} while not reducing to C_H .*

Proof. Let x be a submitted extra-Higgs identity. If x fails to preserve D_H or H_{role} , it is a target update or failed identity claim. If it preserves D_H and H_{role} and is role-equivalent to C_H , Paper 2 classifies it as the same Higgs identity under redescription, not an extra identity. If it uses scale or vacuum success to avoid this reduction, Papers 3 and 4 route those claims into scale or vacuum status rather than identity. The only remaining possibility is a counterexample submission that preserves the entire inherited package while refusing reduction to C_H . Such a submission is a counterexample-burden state, not a standing disposition. It is not supplied by scalarity, vev participation, mixing, coupling, compositeness, or phenomenological success alone. $\square$

Corollary 16.3 (Primitive extra-Higgs deletion).

$$\text{Surv}_{D_H}^{\text{prim}}(\text{ExtraHiggs}_H) = 0$$

for every claim-form in which extra-Higgs identity is grounded only in scalar similarity, vev participation, coupling, mixing, mass scale, vacuum effect, thermal effect, hierarchy repair, or composite presentation. A counterexample-burden submission is not a survivor of this primitive claim-form.

17 Faithful same-scope counterexample burden

A serious objection must preserve the same target rather than enlarge it quietly. The burden is explicit.

Criterion 17.1 (Faithful same-scope extension counterexample). A discharged faithful same-scope counterexample to the same-scope extra-Higgs burden theorem must supply x such that:

1. x preserves D_H and the admissibility-relevant invariant bundle;
2. x preserves H_{role} without role collapse;
3. x preserves identity-redescription discipline R_H^{id} ;
4. scale claims made by x factor through R_H^{scale} ;
5. vacuum or thermal claims made by x factor through R_H^{vac} ;
6. x is not merely representative duplication, enrichment, additional field content, or target update;
7. x does not reduce to C_H under $\sim_H^{\text{id}}$.

Proposition 17.2 (Counterexample burden classification). *A construction failing any item of the faithful counterexample burden is classified as same carrier, enrichment, additional field, role-split reassembly, target update, or failed primitive identity. A construction satisfying the submission form but not the full burden is recorded as $\rho_{\text{burden}}^{\text{ext}}(x) = \text{CounterexampleSubmissionPending}$, not as a standing survivor.*

Proof. Each burden item corresponds to an inherited module. Failure to preserve the domain or invariant bundle is target update. Failure to preserve H_{role} is role failure. Failure of identity redescription reduces to representative duplication or failed primitive. Failure of scale or vacuum route discipline is scale/vacuum standing-smuggling. If the object reduces to C_H , it is same carrier. If the object preserves enough structure to assert a replacement but has not discharged all items, the only faithful classification is counterexample-burden submission. None of these cases is a same-scope standing survivor outside C_H . $\square$

18 Endpoint theorem set

The Paper 5 endpoint is classificatory.

Theorem 18.1 (Paper 5 endpoint theorem set). *The following results hold in the fixed Higgs-domain extension target.*

1. *Every submitted extension claim is incomplete, same-scope classified through $\text{NF}_{H,\text{ext}}^{\text{same}}$, target-updating through $\text{NF}_{H,\text{ext}}^{\text{update}}$, or recorded as a counterexample-burden submission.*
2. *Scalar-like presentation does not imply Higgs identity.*
3. *Carrier preservation yields carrier-indexed enrichment or same-carrier representation, not a new primitive identity.*
4. *Role splitting reassembles into C_H , updates the target, or fails identity continuity.*
5. *Primitive extra-Higgs identity has no same-scope survivor outside C_H absent a fully discharged faithful same-scope counterexample burden.*

Proof. The first line is the separated same-scope/update normal-form necessity theorem together with the deterministic decision procedure. The second is the extra-scalar non-identity theorem. The third is carrier-preserving enrichment. The fourth is role-splitting exhaustion. The fifth is the counterexample-burden theorem and its primitive deletion corollary. Together these close Paper 5’s contribution to the Higgs arc. $\square$

19 Claim and nonclaim register

Table 8: Paper 5 claim and nonclaim register.

ID	Statement	Status
E5-T1	Paper 5 works downstream of the Paper 1–4 Higgs carrier package.	dependency lock
E5-T2	Same-scope extension standing and richer-scope target update require separate normalizers $\mathbf{NF}_{H,\text{ext}}^{\text{same}}$ and $\mathbf{NF}_{H,\text{ext}}^{\text{update}}$.	theorem
E5-T3	Every submitted extension claim receives deterministic layered classification by raw type, boundary/update decision, carrier relation, standing disposition, and where applicable counterexample burden.	classifier theorem
E5-T4	Scalarity, vev participation, coupling, mixing, vacuum effect, thermal role, and phenomenological success do not create primitive Higgs identity.	theorem
E5-T5	Carrier-preserving enrichment remains indexed to C_H .	theorem
E5-T6	Composite and pNGB claims are representation, target update, or primitive failure relative to D_H .	theorem
E5-T7	Role splitting reassembles to C_H , updates target, or fails.	theorem
E5-T8	Unknown Higgs attributes are classified by standing role rather than numerically predicted.	theorem
E5-T9	Faithful counterexample submissions are burden states, not same-scope standing survivors until the full burden is met.	theorem
E5-T10	No same-scope extra-Higgs identity survives outside C_H in the fixed target absent a fully discharged faithful same-domain counterexample.	main endpoint theorem
E5-N1	The paper does not rule out BSM scalar sectors.	nonclaim
E5-N2	The paper does not compute masses, widths, cross sections, couplings, or collider signatures.	nonclaim
E5-N3	The paper does not decide whether any particular BSM model is physically realized.	nonclaim
E5-N4	The paper does not deny that extensions can affect vacuum stability, hierarchy, thermal history, or baryogenesis.	nonclaim
E5-N5	The paper does not reopen Papers 1–4; it works downstream of their carrier, identity, scale, and vacuum modules.	nonclaim

ID	Statement	Status
E5-N6	The paper does not identify target updates as failures; it records their richer-scope burden.	nonclaim

20 Denial obligations

A same-scope objection to Paper 5 must do at least one of the following:

1. break the inherited carrier package from Paper 1;
2. break the identity endpoint from Paper 2;
3. supply a primitive scale ground contrary to Paper 3;
4. supply primitive vacuum or thermal identity contrary to Paper 4;
5. show that scalarity, vev participation, coupling, mixing, or compositeness supplies H_{role} without C_H ;
6. produce a discharged faithful same-scope extension counterexample satisfying the stated burden;
7. show a further extension standing function that changes Higgs-domain standing while preserving all registered fixed-domain loci.

Absent one of these results, the objection either changes target, supplies enrichment, adds a non-Higgs field, duplicates a representative, reassembles into C_H , or fails as primitive identity standing.

21 Conclusion: handoff to synthesis

Paper 5 closes the extension classifier. The conclusion is deliberately limited and structurally strong. Additional scalar, composite, portal, supersymmetric, threshold, and role-split structures are not rejected; they are typed. A submitted extension either preserves the Higgs carrier, enriches it downstream, duplicates a representative, distributes and reassembles its roles, adds non-Higgs field content, changes scope, enters a counterexample-burden register, or fails as primitive identity standing.

The final synthesis receives a closed local package: Paper 1 supplies the carrier; Paper 2 supplies identity; Paper 3 supplies scale standing; Paper 4 supplies vacuum route status; Paper 5 supplies extension classification. The remaining domain-level endpoint is the exhaustion of residual primitive Higgs attributes outside the carrier architecture and its licensed representatives, routes, projections, and target updates.

A Notation glossary

Symbol	Meaning
D_H	fixed same-scope Higgs-domain regime inherited from Paper 1.
C_H	Higgs bridge/fixation carrier.

Symbol	Meaning
H_{role}	role bundle: gauge inheritance, deformation mediation, mass-response readiness, identity continuity, standing transfer.
R_H^{ext}	extension normal form.
Σ_{ext}	added sector, field, carrier, source, or threshold content.
Π_C	carrier-preservation map.
Π_{id}	identity-redescription map.
Π_{scale}	scale-status map relative to Paper 3.
Π_{vac}	vacuum-route-status map relative to Paper 4.
Cert_{ext}	extension standing certificate.
ρ_H^{ext}	layered extension classifier.
$\rho_{\text{burden}}^{\text{ext}}$	counterexample-burden classifier for non-reducing same-scope replacement submissions.
$\text{NF}_{H,\text{ext}}^{\text{same}}$	same-scope extension normalizer.
$\text{NF}_{H,\text{ext}}^{\text{update}}$	target-update extension normalizer.

B Extension outcome matrix: audit form

Outcome	Trigger	Certificate	Nonclaim
Carrier-indexed extension	$\text{Pres}_C(x)$ and inherited modules preserved	$\Pi_C, \Pi_{\text{id}}, \Pi_{\text{scale}}, \Pi_{\text{vac}}$ as needed	no new identity
Additional non-Higgs field	field exists without H_{role} occupation	field/sector declaration	not dismissed physically
Richer-scope update	invariant bundle or carrier inventory changes	target-update certificate	not same-scope counterexample
Role-split reassembly	distributed functions preserve H_{role}	reassembly map	not multiple identities
Failed extra-Higgs identity	primitive standing asserted from representative feature	none closes	failure of identity claim only
Counterexample-burden submission	same-scope preservation with non-reduction to C_H is asserted	full counterexample burden	burden state, not standing survivor

C Model-neutral standing tests

1. Does the extension preserve D_H or declare $D_{H,+}$?
2. Does it preserve C_H as identity carrier?

3. Does it preserve every component of H_{role} ?
4. Are scale claims routed through R_H^{scale} ?
5. Are vacuum and thermal claims routed through R_H^{vac} ?
6. Does the claim rely on scalarity, vev participation, mixing, coupling, or success as primitive standing?
7. If role splitting is present, does it reassemble into C_H or declare target update?
8. If a replacement is claimed, is it recorded as a counterexample burden or discharged by the faithful counterexample certificate?

D Example classifications for common extension families

The classifications below are schematic audit forms. They are not verdicts on specific models.

Family	Conservative classification	Stronger burden if identity is claimed
Singlet portal	carrier-preserving enrichment or additional non-Higgs field	show occupation or preservation of H_{role}
Two-Higgs-doublet presentation	role split, enrichment, or target update	show same-scope reassembly or discharge the faithful counterexample burden
Composite Higgs	same-carrier representation or target update	supply identity-redescription map
pNGB Higgs	protection/enrichment or target update	show protection does not replace carrier with scope shift
Supersymmetric Higgs sector	richer-scope update unless same-scope preservation is certified	specify carrier inventory and descent
Radion/dilaton-like scalar	scale projection, additional field, or target update	show scale role is not primitive identity
Threshold/UV completion	target update or scale route	supply invariant-bundle and descent certificate

E Source support map

Source	Imported role	Use in Paper 5
Kernel/admissibility support	fixed-domain discipline, AMetric boundary, no repair, no transfer	classifier closure and no hidden authorizer
Paper 1	carrier package ($D_H, C_H, H_{\text{role}}, R_H, \mathbf{SF}_H$)	base target and carrier relation
Paper 2	identity endpoint $\text{Id}_H = \text{Occ}(C_H)$	no extra identity from representatives

Source	Imported role	Use in Paper 5
Paper 3	no primitive Higgs scale	scale import discipline
Paper 4	vacuum route status	vacuum/thermal import discipline
Arc plan	Paper 5 role in six-paper program	skeleton and guardrails

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